Weil descent beyond covers: four angles, measured (round 3)

Status. A research search under a protocol frozen before every run (`PROTOCOL.md`: Addendum A and Amendments B–G, each committed before the runs it governs). Each item has a recorded run, a run report with a graph, and checks. Nothing here is a speed record on a curve anyone deploys. Every cost is a counted operation, never wall-clock. Conductor is not installable on this host (installer HTTP 403 through the session proxy).

Tags: **[E]** exact enumeration or counted measurement in this round; **[T]** proved here; **[L]** recalled from the literature and not re-read this round (novelty audit owed).

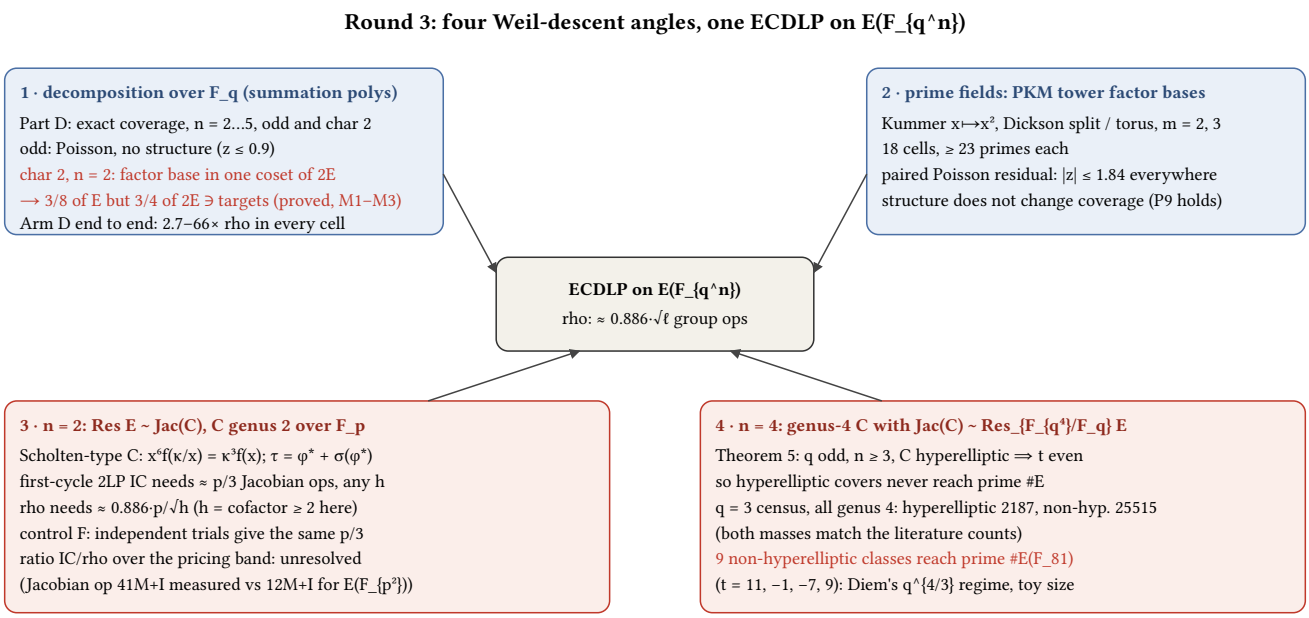


Figure 1: Round 3 at a glance

Answer

The four angles are (a) decomposition over `F_{q^n}` with summation polynomials and Weil descent; (b) prime fields through Petit–Kosters–Messeng (PKM) descent onto a subspace; (c) the constant factor at `n = 2`, from genus-2 index calculus over `F_p`; (d) genus-4 Jacobians isogenous to `Res_{F_{q^4}/F_q} E`. They are answered in order below.

1. **Decomposition coverage is Poisson, with one structural exception [E, T].** For generic `E/F_{q^n}` and the factor base `x ∈ F_q`, the coverage (the share of points that are a sum of `n` factor-base points) was computed exactly over the whole group. In odd characteristic at `n = 2…5` it matches a random `x`-set of the same size: the paired residual from Poisson has `|z| ≤ 0.94` (§1.1).

   The exception is **binary `n = 2`**, where `π → 3/8` rather than the Poisson value `1 − e^{−1/2} = 0.3935` (z = −4.72). The cause is proved (§1.2):

   - every `x ∈ F_q` has `Tr_{F_{q²}/F_2}(x) = 0`, so the whole factor base lies in one coset of `2E`;
   - the descended relation map is F_2-affine, and its image lies in the hyperplane `Tr(w/x_3²) = 0`;
   - so a point decomposes only if `Tr(x_3) = Tr(a)`, which holds for half the group.

   Inside that half the multiplicities are exactly `Bin(2, 1/2)`. The identity and the exact reconstruction of coverage from F_2 cosets agree on all 21 curves.

   Attack reading, corrected after the first push: the earlier text called the effect neutral to slightly negative, measured over the whole group. In an attack every target `aP + bQ` lies in the odd ℓ-subgroup, which sits inside `2E`, so the relevant figure is coverage of `2E`. That is exactly 3/4, against 0.3935 for the Poisson model at the same factor-base size: about 1.9× more relations per target. This is a correction of the cost model (accounting), not a new algorithm. Coverage of the ℓ-subgroup itself was not measured separately.

   End to end at `n = 2`, the decomposition route (Arm D) costs **2.7–66× rho in every pricing cell**. Its oracle needs a quartic root-finding, about 1,000 M per call (§3).

2. **Prime fields via PKM: no coverage effect [E].** The Kummer (`x ↦ x²`) and Dickson (split and torus) tower factor bases were tested on 18 cells with at least 23 primes each. Their exact coverage is indistinguishable from random `x`-sets: every paired-residual `|z| ≤ 1.84`. The algebraic structure of the tower does not create extra relations (P9 holds), so any PKM advantage would have to come from solving, not from coverage (§2).

3. **`n = 2`, genus-2 index calculus against rho: a constant-factor race, unresolved [E].** The route uses a Scholten-type `C/F_p` with `Jac(C) ~ Res_{F_{p²}/F_p} E` and the verified transfer `τ`.

   - **Operation counts.** First-cycle double-large-prime index calculus needs about `p/3` Jacobian operations (0.37 at cofactor `h = 4`, 0.27 at `h = 8`). Rho needs `0.886·p/√h` (0.44 and 0.31). A control with independent trials gives the same counts, so this is not rho hiding inside the walk.
   - **Cost per operation.** This tool's genus-2 addition costs 41 M + I, against 12 M + I for an `E(F_{p²})` addition.
   - **Ranking.** On 1,260 verified runs up to `log2 #E = 40`, Arm J's cost ratio to rho spans the pricing band (0.51–2.83), so the rule ranks it **unresolved** at every size. At 40 bits rho is cheaper in 10 of 12 cells.
   - **Distance to rho.** About 2.8× under measured costs at `I = 3M`. Every curve in this family has `h ≥ 2`, and prime-order `E` (`h = 1`) is where rho loses most; that is the next experiment (§7).

4. **`n = 4`: only non-hyperelliptic genus-4 curves reach prime-order `E` [T, E].** Theorem 5 (round 3, proved): for `q` odd and `n ≥ 3`, a hyperelliptic genus-`n` curve `C` with `Jac(C) ~ Res_{F_{q^n}/F_q} E` forces the trace `t` even. That makes `#E` even, so hyperelliptic covers never reach prime-order `E`.

   The exhaustive census of every genus-4 curve over `F_3` confirms this and finds the way around it:

- **Masses.** The hyperelliptic mass is 2,187 (`= 3^7`). The non-hyperelliptic mass is 25,515 (`= q^9 + q^8 − q^6`), both matching the literature counts.
- **Hyperelliptic.** All 1,100 Res-shaped hyperelliptic models have even `t`.
- **Non-hyperelliptic.** Of the 158 Res-shaped non-hyperelliptic classes, **9 reach a prime-order `E/F_81`**: `t ∈ {11, −1, −7, 9}`, `#E ∈ {71, 83, 89, 73}`.

Those are the curves where Diem's plane-curve index calculus, `q^{4/3}`, would apply instead of rho's `q²`. The result is toy-sized: the census shows existence; it does not construct the map (§4).

1. Item 1: decomposition over `F_{q^n}` (Parts D and Arm D)

1.1 Exact coverage (Part D, run `item1D-run-20261018T2056Z`) [E]

Setup. `E : y² = x³ + ax + b` (odd) or `y² + xy = x³ + ax² + b` (binary) over `F_{q^n}`, with coefficients outside every proper subfield, 3 curves per field.

Arms.

- **subfield:** `F = {P : x(P) ∈ F_q}`;
- **null:** a random `x`-set of equal size;
- **positive:** an interval `{±jG}`.

Statistic. For each arm, the exact number of distinct `n`-fold multiset sums over the whole group. The primary statistic (Amendment B) is each arm's residual `π − (1 − e^{−λ})`, with `λ` its multiset count over `#E`. It is paired per curve, subfield against null.

field	n	curves	λ	π subfield	π null	resid subfield	resid null	diff	z
odd	2	672	0.503	0.3949	0.3931	+0.00017	+0.00001	+0.00016	0.94
odd	3	63	0.186	0.1576	0.1787	−0.00985	−0.00977	−0.00007	−0.10
odd	4	21	0.083	0.0605	0.0625	−0.01779	−0.01691	−0.00088	−0.19
odd	5	9	0.037	0.0210	0.0187	−0.01439	−0.01174	−0.00265	−0.43
2	2	21	0.507	0.3730	0.4054	−0.02465	+0.00615	**−0.03080**	**−4.72**
2	3	15	0.168	0.1385	0.2039	−0.01150	−0.02743	+0.01592	1.96
2	4	9	0.147	0.0818	0.0711	−0.03826	−0.02154	−0.01672	−1.06
2	5	9	0.049	0.0265	0.0529	−0.02066	−0.04763	+0.02696	1.11

The positive control falls far below Poisson in every cell, with π between 0.003 and 0.03 against 0.04–0.4. So the statistic sees structure when structure is there. The Gaudry heuristic `1/n!` is not the coverage: π follows Poisson at the measured λ, as P7 predicted.

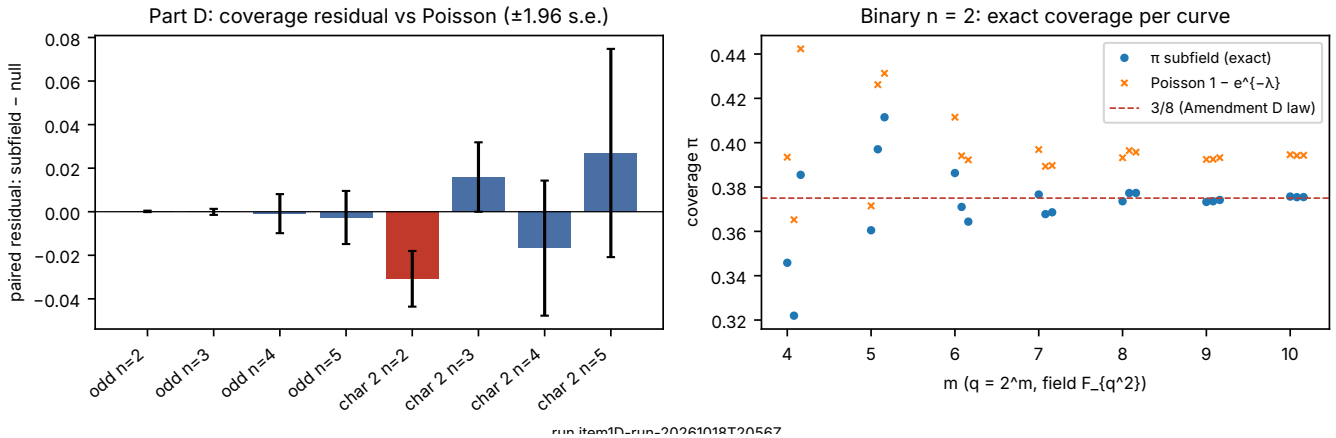


Figure 2: Part D

1.2 The binary `n = 2` deficit (Amendment D, run `item1D-mult-run-20261018T2106Z`) [T, E]

Take `y² + xy = x³ + ax² + b` over `F_{q²}` with `q = 2^m`, and write `e_1 = x_1 + x_2`, `e_2 = x_1x_2`. The descended relation equation is

`S_3 = x_3² e_1² + e_2² + x_3 e_2 + b = 0`.

Squaring is additive in characteristic 2, so `L_{x_3}(e_1, e_2) = x_3² e_1² + e_2² + x_3 e_2` is F_2-linear from `F_q²` (dimension `2m`) to `F_{q²}` (dimension `2m`).

Lemma. `Tr(L(e)/x_3²) = 0` for every `e ∈ F_q²`.

Proof. Write `T = Tr_{F_{q²}/F_2}`. Then `T(L/x_3²) = T(e_1²) + T((e_2/x_3)² + e_2/x_3) = T(e_1) = 0`. The middle term vanishes because `T(z² + z) = 0`. The last step holds because `e_1 ∈ F_q`, so `T(e_1) = Tr_{F_q}(2e_1) = 0`. ∎

Hence `rank L ≤ 2m − 1`. A target is solvable only if `T(b/x_3²) = 0`. A point `(x_3, y_3)` of `E` satisfies `T(x_3 + a + b/x_3²) = 0`, so a decomposable point has `T(x_3) = T(a)`. That condition is the classical halving criterion for `2E`. Every factor-base point has `T(x) = 0` and lies in one coset of `2E`, so the 2-sums lie in `2E`, which is half the group.

Measured on 21 curves, `m = 4…10`:

- **M1.** `x(P+Q) + x(P−Q) = x_1x_2/(x_1+x_2)²`, independent of `a` and `b`, holds on every pair (0 violations).
- **M2.** `dim ker L = 1` for every `x_3 ∉ F_q`. Exactly half of these are solvable (523,776 of 1,048,576 at `m = 10`). Every solution coset has size 2 (0 sizes that are not a power of 2).
- **M3.** Rebuilding coverage from the cosets, after the split and point-existence conditions, gives the measured distinct count exactly on all 21 curves (0 mismatches).

The point multiplicity histogram at `m = 10` is `(0, 1, 2, ≥3) = (0.626, 0.250, 0.125, 0)`. Inside 2E that is exactly `Bin(2, 1/2)`, so `π = (1/2)(3/4) = 3/8`. The measured π at `m = 8…10` is `0.3737–0.3772`.

Prediction P7b fails as worded. It said the index of dispersion would exceed 1. The measured dispersion is `1.0007 ± 0.030`, against `0.9725 ± 0.032` for the odd subfield. The deficit is a support effect (half the group is unreachable, and inside it the multiplicities are under-dispersed), not clustering.

The odd `n = 2` subfield multiplicities also stop at 4 (Bézout on the descended quartic). At `λ ≈ 0.5` that truncation has no visible effect.

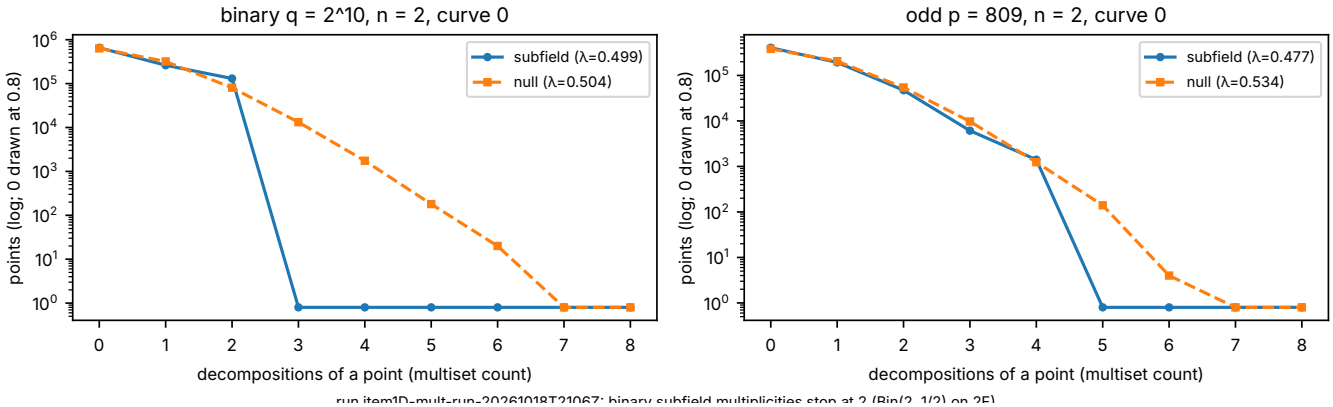


Figure 3: Multiplicities

Attack reading (corrected after the first push). Over the whole group the yield is 3/8 against the Poisson value 0.3935, which the first version of this report read as "4.7% fewer relations per target". That used the wrong denominator. Every attack target `aP + bQ` lies in the odd ℓ-subgroup, which is contained in 2E. Inside 2E the multiplicities are exactly `Bin(2, 1/2)`, so coverage of 2E is **3/4**. A random factor base of the same size reaches 0.39 there, so the subfield factor base yields about **1.9× more relations per target**, or 1.5× against Gaudry's `1/n! = 1/2` estimate.

The gain is bounded by the index of the coset subgroup, which is at most the 2-part of the cofactor. It is a correction to the cost model of binary `n = 2` decomposition (accounting), not a new algorithm. Coverage of the ℓ-subgroup itself was not measured separately; that is the first follow-up experiment. Classification: an exact structural fact plus a cost model correction, not an improvement. We have not checked whether it is in the literature (novelty audit owed); the 2E coset observation may be folklore.

2. Item 2: prime fields through PKM factor bases (Part P, run `item2P-run-20261018T2056Z`) [E]

Setup. Prime-order `E/F_p`, with `V` the `2^t` roots of a Kummer tower (`x ↦ x²`) or a Dickson tower (`x ↦ x² − 2`), split or torus, and `t` at the sizing `|V|^m ≈ m!·r` and one step either side. The coverage of `m`-fold sums is exact over the whole group. There are 18 cells with 23–24 primes each; Amendment B's paired residual is the statistic.

family	m	offsets −1 / 0 / +1: paired residual difference (z)
Kummer	2	−0.00024 (−1.10) / −0.00041 (−1.40) / −0.00024 (−1.84)
Kummer	3	−0.00046 (−0.43) / +0.00043 (0.26) / +0.00043 (1.36)
Dickson split	2	+0.00008 (0.34) / +0.00015 (0.39) / −0.00004 (−0.34)
Dickson split	3	−0.00053 (−0.85) / +0.00236 (1.71) / −0.00008 (−0.22)
Dickson torus	2	−0.00013 (−0.53) / −0.00005 (−0.23) / −0.00005 (−0.45)
Dickson torus	3	−0.00014 (−0.16) / −0.00100 (−0.48) / +0.00018 (0.71)

There is no cell with `|z| > 2`. The three Kummer `m = 2` cells lean negative (z −1.1, −1.4, −1.8). That is within spread, and it is recorded as a lean, not an effect. P9 holds. Coverage at the sizing point is 0.58–0.65 (`m = 2`) and 0.59–0.61 (`m = 3`), the Poisson values.

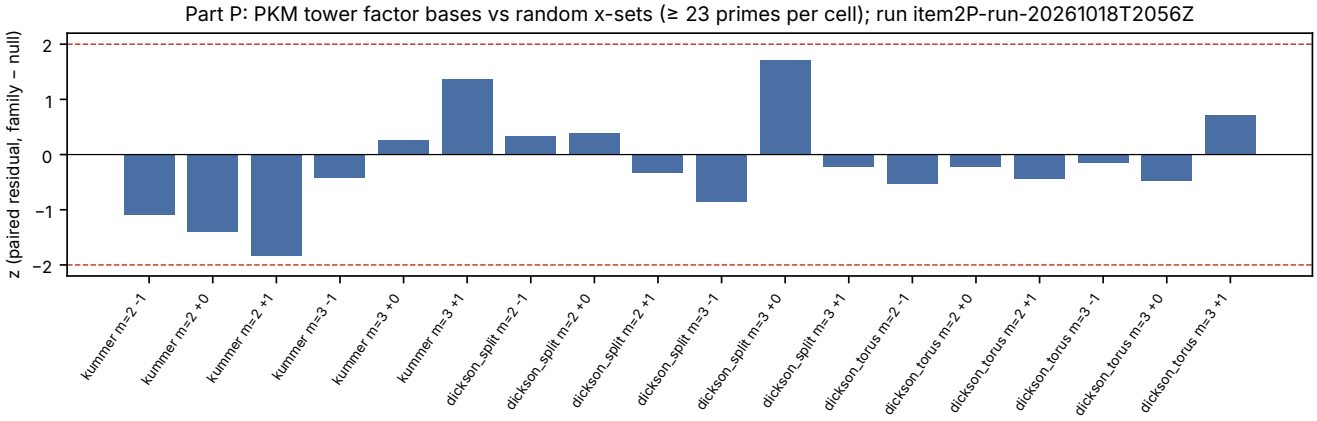


Figure 4: Part P

3. Item 3: the `n = 2` routes end to end (Parts O/J; Amendments C, E, F, G)

3.1 Construction and checks [E]

The curve. `C : y² = f(x)` over `F_p`, with `f = (x − r)(x − κ/r)·g`, `g` symmetric of degree 4, `x⁶f(κ/x) = κ³f(x)`, and `κ` a non-square.

The map. With `s² = κ`, the map `φ(x, y) = (x + κ/x, y(x + s)/x²)` sends `C` to `E : v² = h(u)(u + 2s)` over `F_{p²}`. Its Weierstrass form is `Y² = X³ + h_1X² + h_0h_2X + h_0²f_6`, with `h_i` the Taylor coefficients of `h` at `u_0 = −2s`.

The transfer. `τ(P) = φ^*P + (φ^*P)^σ` lands in the quintic model of `C`, obtained by sending the rational Weierstrass point `x = r` to infinity.

Checks (run `itemOJ-run-20261018T2124Z/checks.txt`). On 6 instances (p = 263 … 4507):

- `#C(F_p) = p + 1`, which is the Res-shape;
- `ℓ·P = O`;
- `τ` is a valid divisor and a homomorphism (`τ(P + P') = τP + τP'`);
- `τ(P) ≠ 0` and `ℓ·τ(P) = 0`;
- `N·D = 0` for random `D ∈ Jac(C)(F_p)`.

Every one holds.

The decomposition oracle (two descended equations, then a quartic in `e_1`) equals brute force on all 6,068 points of 4 instances with `p ≤ 101`.

Per-operation costs, measured (K5).

operation	cost
affine `E(F_{p²})` addition	12 M + 1 I
genus-2 addition, this tool's reduced formulas	41 M + 1 I
genus-2 doubling, reduced formulas	48 M + 1 I
generic Cantor addition	113 M + 4 I

The reduced formulas agree with Cantor on every tested pair.

3.2 What wins the race: the large-prime graph, not the factor base [E]

Every IC answer arrived as a direct relation `a + bk = 0`: a cycle among large primes with no factor-base column. It came long before `|B| + 10` B-relations existed. So the factor-base size does not matter: `c = 1, 2, 4` give the same costs within spread.

In a random multigraph on `n ≈ p/2` vertices (the degree-1 points up to sign), the expected number of edges before the first cycle is `∫_0^{1/2} √(1 − 2c) dc · n = n/3`. Half the trials split, so this predicts `p/3` Jacobian operations (P8b). Rho with the negation map needs `0.886·√ℓ ≈ 0.886·p/√h`.

Control F (run `itemF-control-run-20261018T2128Z`). Independent random trials (Jind, no walk) need the same number of operations as Arm J's walk:

arm	h	runs	trials/p	s.e.
Jind (independent trials)	4	33	0.369	0.017
Jind	8	21	0.272	0.023
Jwalk (Arm J's walk)	4	33	0.344	0.023
Jwalk	8	21	0.292	0.030

So the count is not rho hiding inside the walk (P8c holds). The count does depend on the cofactor: `h = 8` needs fewer trials than `h = 4`. Relations land in the `ℓ`-subgroup, so for a given point only `1/h` of partners are possible, which raises the rate of repeated edges. The simple `p/3` law describes `h ≈ 4–8` only roughly.

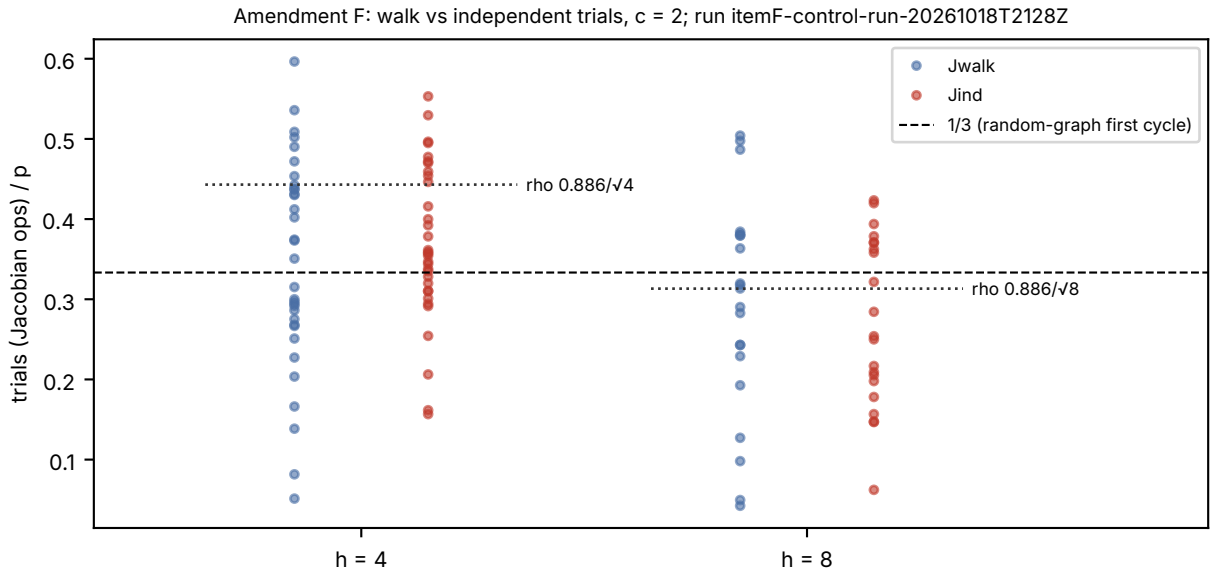


Figure 5: Control F

3.3 End-to-end costs (run `itemOJ-run-20261018T2132Z`, recorded result) [E]

Scale of the run.

- 6 sizes, `log2 #E = 20 … 40` (`p ≈ 2^10 … 2^20`), 3 instances each; cofactors `h ∈ {4, 8}`.
- 10 targets per instance, so 30 runs per arm and size.
- Arms: R (rho); J and D at `c = 1, 2, 4`.
- **All 1,260 answers verified (`k·P = Q`).**

Reference arm (K6). Mean rho `steps/p` against `0.886·√ℓ/p`:

log2 #E	rho, measured	theory
20	0.327 ± 0.029	0.357
24	0.346 ± 0.038	0.400
28	0.429 ± 0.044	0.400
32	0.361 ± 0.048	0.357
36	0.499 ± 0.054	0.443
40	0.388 ± 0.034	0.400

Pricing band (Amendment C). Each cell fixes one value of each:

- `I ∈ {3M, 2·log2(p)·M}`;
- the Jacobian price: measured, or one `E(F_{p²})` addition;
- the native weight: `0`, `0.1` or `1`.

That gives 12 cells for Arm J and 6 for Arm D. An arm is ranked against rho only where the order holds in every cell. The table gives the IC/rho cost ratio over the band at `c = 2`:

log2 #E	Arm J: min–max	cells IC cheaper	Arm D: min–max	verdict J / D
20	0.51–1.53	6/12	2.73–3.83	unresolved / rho cheaper
24	0.64–1.84	6/12	7.36–12.2	unresolved / rho cheaper
28	0.73–2.12	6/12	13.9–27.0	unresolved / rho cheaper
32	0.84–2.47	5/12	22.6–49.5	unresolved / rho cheaper
36	0.75–2.22	5/12	21.0–51.3	unresolved / rho cheaper
40	0.95–2.83	2/12	24.8–66.3	unresolved / rho cheaper

`c = 1` and `c = 4` are within a few percent of `c = 2`. The best `c` per size is a post-hoc minimum, which favours IC, and changes no verdict (`ranking.csv`).

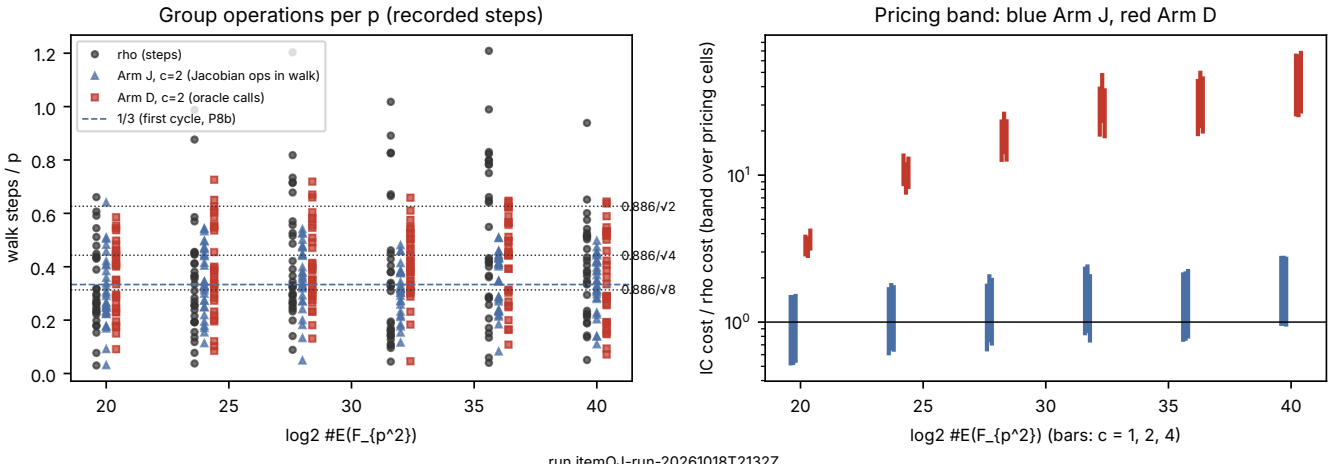


Figure 6: Routes

Reading.

- **Arm D, the decomposition route.** Rho is cheaper in every cell at every size, and the ratio grows with `log p`. Each oracle call computes `x^p` modulo the descended quartic, about 1,000 M at `p ≈ 2^20`, against rho's 15 M per step. P8 said Arm D's per-trial cost would be below Arm J's; that is **falsified**. The oracle costs about 20 Jacobian operations.
- **Arm J, the genus-2 route.** It is unresolved at every size, and drifts toward rho as the fixed setup (`r` scalar multiplications per arm) amortises. At 40 bits rho is cheaper in 10 of 12 cells. Under the measured Jacobian cost with `I = 3M`, the ratio is 2.80–2.83. Under the IC-favourable price (one `E(F_{p²})` addition per Jacobian operation) it is 1.10–1.32, or 0.95–1.02 with Fermat-priced inversion.
- **Where the costs come from.** Asymptotically, `J/rho ≈ (c_J/c_E) · (trials_J/steps_rho)`. The count ratio is `≈ 0.84` at `h = 4` and `≈ 0.9` at `h = 8` (control F). The cost ratio `c_J/c_E` is `(41 + I)/(12 + I) ≈ 2.9` at `I = 3M`.
- **Distance remaining to rho on these curves.** About 2.8× under measured costs. Closing it needs `c_J/c_E ≤ 1.2` at `h = 4`. That is out of reach for genus-2 arithmetic over `F_p` against elliptic arithmetic over `F_{p²}`.
- **Exponent.** Both arms are linear in `p` (P8). The fitted slopes of `log cost` on `log p` (R 0.70, J 0.78) are below 1 because the setup amortises; `steps/p` is flat for both.

Classification. The first measurement of this route: it sets the baseline, so it is not an improvement over an earlier one. The rho repairs (Amendments E and G) are engineering applied to the reference arm, and they make rho stronger. The leaderboard is unchanged: these instances (Scholten-type `E/F_{p²}`, `log2 #E ≤ 40`) are not on it.

4. Item 4: genus-4 curves and `Res_{F_{q^4}/F_q} E` (run `item4-run-20261018T2049Z`) [T, E]

Theorem 5 [T]. Let `q` be odd, `n ≥ 3`, and `C/F_q` hyperelliptic of genus `n` with `Jac(C) ~ Res_{F_{q^n}/F_q} E`. Then `t = tr(E/F_{q^n})` is even.

Proof sketch. The full proof is in the Part T tool comments and the round-3 protocol.

- The Frobenius on `Jac(C)[2]` permutes the `2n + 2` Weierstrass points. With cycle lengths `c_i`, its characteristic polynomial is `∏(T^{c_i} + 1)/(T + 1)²` over F_2.
- The characteristic polynomial of `Res E` reduced mod 2 is `χ̄_E(T^n) = T^{2n} + tT^n + 1`.
- If `t` is odd, this is `Φ_3(T^n) = Φ_3(T^m)^{2^a}` with `n = 2^a·m`, `m` odd. That has a root of order `3m`.
- The only factors `T^c + 1` that contain such a root have `3m | c`. Covering every root then forces `Σ c_i ≥ 3n`. But `Σ c_i = 2n + 2`, which is less than `3n` once `n ≥ 3`. ∎

Part T enumerates cycle types for `n ≤ 12` and finds no odd-`t` solution at `n ≥ 3`. At `n = 2` the odd-`t` cycle types are `[6]` and `[3, 3]`. That is why round 2 found prime-order `E` at `n = 2` and none at `n = 3`.

Census over `F_3`, genus 4, every curve [E].

- **Hyperelliptic (Part H4).** There are 104,976 squarefree degree-10 binary forms; divided by 48 that gives mass 2,187 = `3^7` (K1). 1,100 models are Res-shaped. Every one has even `t`, and every cycle type is a product of powers of 2 (K2: 0 violations).
- **Non-hyperelliptic (Part K).** Canonical models lie on a quadric in $\mathbb{P}^3$, of one of three kinds:

quadric	forms	smooth	mass	Res-shaped forms	classes
split $\mathbb{P}^1\times\mathbb{P}^1$, (3,3) forms, `|G| = 1152`	43,046,720	21,830,400	9,475	81,216	36
non-split, `|G| = 1440`	43,046,720	27,299,520	9,479	110,880	39
cone `P(1,1,2)`, `|G| = 1296`	14,348,906	8,503,056	6,561	101,520	83
total			**25,515**		**158**

  The total mass equals `q^9 + q^8 − q^6 = 25,515`, the literature count of non-hyperelliptic genus-4 curves over `F_3` (K3, [L]). Orbit integrity holds: K4 and K5 have 0 violations.
- **Prediction P6 holds.** Nine Res-shaped non-hyperelliptic classes have an elliptic trace that exists and `#E(F_81)` prime:

quadric	t	#E

split	11	71
non-split	−1	83
cone	−7	89 (×3)
cone	9	73 (×3, one with automorphism group of order 3)
cone	−1	83

Odd traces are common among non-hyperelliptic classes (see the histogram). A further 32 classes have a trace with no elliptic curve, because `3 | t` and `t ∉ {0, ±9, ±18}`.

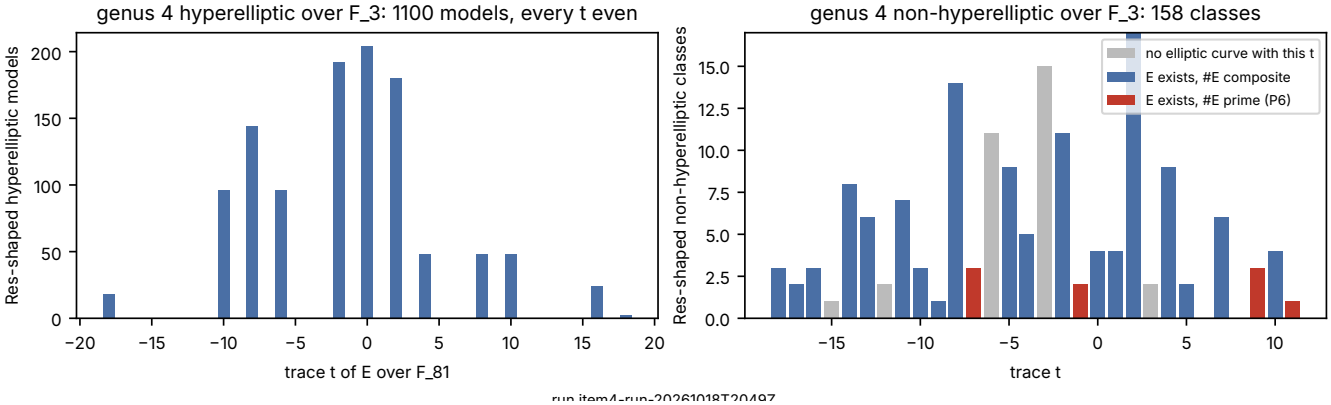


Figure 7: Item 4

Reading.

- **Costs at `n = 4` (exponents, [L]).** Rho on `E(F_{q^4})` costs `q²`; Gaudry decomposition and hyperelliptic genus-4 index calculus cost `q^{3/2}`; Diem's non-hyperelliptic plane-quintic route costs `q^{4/3}`.
- **What the theorem and census establish.** The lower-exponent covers that would reach prime-order `E` must be non-hyperelliptic, and such classes exist (9 of 158 at `q = 3`).
- **What they do not establish.** That such a `C` can be found for a given `E` at cryptographic size, or that the map `C → E` can be computed. By Tate's isogeny theorem, a non-constant map `C → E` over `F_{q^4}` exists once the isogeny does, but its degree is unknown. This is an existence result at toy size, not an attack.

5. Checks and controls

check	where	result
K1 hyperelliptic genus-4 mass `= q^7`	item 4	2187 = 2187
K2 hyperelliptic Res-shaped: `t` even, cycle types powers of 2	item 4	0 violations / 1100
K3 non-hyperelliptic mass `= q^9 + q^8 − q^6`	item 4	25515 = 25515
K4, K5 orbit and stabiliser integrity	item 4	0, 0
Part D positive control below Poisson	item 1	every cell, π 0.003–0.03
M1–M3 (Amendment D)	item 1	0 / 0 / 0 violations, 21 curves
Part P positive control below Poisson	item 2	every cell
K1 decomposition oracle = brute force (`p ≤ 101`)	item 3	6,068 points, 0 mismatches
K2–K4: `#C(F_p) = p + 1`, `τ` a homomorphism, `τ(P) ≠ 0`, `ℓ·τ(P) = 0`, `N·D = 0`	item 3	all true on 6 instances
K5 fast genus-2 formulas = Cantor	item 3	agree on 400 pairs per size
K6 rho mean steps/p vs `0.886·√ℓ/p`	item 3	see §3
every answer `k·P = Q` verified	item 3	all runs
unit tests	tools	genus4 4, pkm 3, decomposition 5, n2_routes 5: all pass

6. What is not established

- **Item 1.** Arm D's oracle cost uses gcd with `x^p − x` on a quartic. A cheaper root-finding method would lower Arm D's constant but not its exponent. The 2E coset fact is proved only for binary `n = 2`. Its analogue for even `n ≥ 4` (sums of `n` points from one coset) is predicted but was not separated from noise here: binary `n = 4` gives z −1.06 on 9 curves.
- **Item 2.** Only coverage was measured. The PKM solving cost (the algebraic systems over the tower) was not.
- **Item 3.** One family (Scholten-type, rational 2-torsion, so `h ≥ 2`). Sizes go only to `log2 #E = 40`. The Jacobian cost is this tool's formula (41 M + I), not an optimised one. See §3 for the scope of the ranking.
- **Item 4.** `q = 3` only. Existence, not construction. No timing.
- **Literature.** No source was re-read this round. Every [L] claim and every "not in the literature" needs the novelty audit.

7. Next improvements

In order of expected value:

1. **Prime-order `E` with a genus-2 Weil-restriction cover (`h = 1`).** The cofactor costs rho √h, and it costs Arm J much less (control F). With `h = 1` rho needs `0.886·p`, while Arm J's count is not yet measured at `h = 1`.
   - Every instance here has `h ≥ 2`, because the rational Weierstrass point `x = r` (needed for the quintic model) puts 2-torsion on `E`.
   - Next step: build `E` without rational 2-torsion. That needs real-model (sextic) genus-2 arithmetic or a different Scholten family, then measure Arm J's trials per `p` at `h = 1`.
   - If they stay near `0.4·p`, the operation-count advantage is about 2.2×. That would beat a cost ratio `c_J/c_E` of about 2.
2. **Cheaper genus-2 arithmetic, measured.** Implement and measure an explicit-formula addition with fewer than 41 M; the literature reports about 25 M + I, but that figure is not admissible until it is measured here. Batch the inversions across parallel walks, for both arms.
3. **Item 1, even `n ≥ 4` binary.** Test the `2E`-coset confinement at `n = 4` with enough curves, and decide whether the free trace filter changes the decomposition cost at `n = 4`, where the oracle is no longer cheap.
4. **Item 4 at `q = 5`.** Rerun the non-hyperelliptic census, where Part K costs about `q^9`. Then try to construct the map `C → E` for one of the 9 prime-order classes at `q = 3` (the degree of the cover), and time Diem's algorithm on it as a toy.
5. **Item 2.** Measure PKM solving cost, not coverage, on the same prime-field cells.

8. Canonical graphs checked

This round adds new graphs only: `figures/item1_coverage`, `item1_multiplicity`, `item2_pkm`, `item4_traces`, `itemOJ_routes`, `itemF_control` and `routes_diagram.svg`. The study-level graphs of rounds 1 and 2 (`research/teske_variants_field_types_20261018/figures/*`, `research/weil_restriction_isogeny_census_20261018/figures/*`) were checked and left unchanged, because no number they plot changed. The index-calculus leaderboard is unchanged because this round touches none of its instances (see §3).

Run reports

- `reports/item4-run-20261018T2049Z/run-report.pdf`
- `reports/item1D-run-20261018T2056Z/run-report.pdf`
- `reports/item1D-mult-run-20261018T2106Z/run-report.pdf`
- `reports/item2P-run-20261018T2056Z/run-report.pdf`
- `reports/itemOJ-run-20261018T2124Z/run-report.pdf` (superseded for rho by Amendment G; kept)
- `reports/itemF-control-run-20261018T2128Z/run-report.pdf`
- `reports/itemOJ-run-20261018T2132Z/run-report.pdf` (the recorded Part O/J result)

References [L, not re-read this round]

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- Petit, Kosters, Messeng, *Algebraic approaches for the elliptic curve discrete logarithm problem over prime fields*, PKC 2016.
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